Internship Report

On

"Empowering Coastal Communities and Strengthening Marine Biodiversity Conservation in Bangladesh: Integrated Approaches for Five Marine Protected Areas and 19 Coastal Districts of Bay of Bengal" and "Coral Restoration and Mapping in Saint Martin's Island"

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Organization Information

Name of the Organization:

Cosmos Foundation

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Overview:

Cosmos Foundation is the non-profit and philanthropic arm of Cosmos Group, dedicated to promoting sustainable development, scientific research, and policy advocacy in Bangladesh and beyond. The organization undertakes multidisciplinary projects in areas such as climate change, marine conservation, renewable energy, peacebuilding, and international affairs. Through research, education, and strategic partnerships, Cosmos Foundation aims to contribute to long term environmental stewardship and social progress.

As part of its marine and environmental initiatives, the Foundation has supported projects related to marine biodiversity, coral reef protection, coastal ecosystem management, and community based conservation planning. These efforts are aligned with national and global goals such as the SDGs and the Paris Climate Agreement.

1. Introduction

1.1 Background

Bangladesh's coastal and marine zones are rich in biodiversity and provide essential ecosystem services and livelihoods for millions living along the Bay of Bengal. However, these ecosystems are increasingly threatened by overfishing, habitat destruction, pollution, and the impacts of climate change (DoE, 2018; MoEFCC, 2021). To conserve marine biodiversity and ensure sustainable use of marine resources, the Government of Bangladesh has declared five Marine Protected Areas (MPAs): **Swatch of No Ground MPA**, **Nijhum Dwip MPA**, **St. Martin's Island MPA**, **Middle South Bay MPA**, and **Teknaf Peninsula MPA** (DoF, 2023). Despite their ecological importance, these MPAs face challenges such as limited enforcement, insufficient scientific research, and weak community participation (IUCN, 2020). Many coastal communities also struggle with poverty and lack access to alternative livelihoods. This project seeks to address these issues through integrated, community-driven conservation efforts across the five MPAs and 19 coastal districts, aiming to protect biodiversity, enhance resilience, and support sustainable development along Bangladesh's coast.

Saint Martin's Island, located in the northeastern Bay of Bengal, is Bangladesh's only coral island and a critical biodiversity hotspot. It supports a unique marine ecosystem, including over 60 species of corals, various reef fish, mollusks, sea turtles, and dolphins. Despite being declared a Marine Protected Area (MPA), the island's fragile coral habitats are rapidly deteriorating due to a combination of natural and human-induced factors. Rising sea surface temperatures have led to coral bleaching, while unregulated tourism, marine pollution, and coastal overdevelopment have caused habitat degradation, increased coral diseases, and a decline in live coral cover. Enforcement of conservation measures remains weak, and scientific monitoring is limited, leaving significant knowledge gaps in reef health and resilience. In this context, coral restoration and mapping efforts are urgently needed to identify vulnerable areas, guide management strategies, and support the long-term sustainability of this ecologically important marine area (IUCN Bangladesh, 2020; Department of Fisheries [DoF], 2023; Rashid et al., 2014).

1.2 Objectives

- Strengthen the Conservation of Marine Megafauna in the Bay of Bengal.
- Improved Coordination, Political Commitment, and Standard Practices of Marine Protected Area Management.
- Enhanced Capacity to safely and efficiently combat pollution and marine crime.
- Enhanced Communication Education and Public awareness about Marine Conservation issues of the Bay of Bengal.

- To assess coral health and vulnerability in Saint Martin's Island using field data.
- To propose a scientifically supported coral restoration and mapping framework.

- To identify community-level and institutional strategies to enhance reef conservation.
- To support the development of a Coral Vulnerability Index (CVI) and coral transplantation plan.

1.2 Purpose of the Internship

The purpose of this internship was to gain hands-on experience in environmental project planning and proposal development focused on marine biodiversity and coastal community sustainability. Through direct involvement in a comprehensive proposal, the internship offered insights into the strategic planning of large-scale conservation initiatives, including integration of ecological, policy, and socio-economic components.

2. Literature Review

2.1 Marine Biodiversity in the Bay of Bengal:

The Bay of Bengal hosts a diverse array of marine species, including critically endangered dolphins, whales, sharks, and sea turtles. Bangladesh's coastal waters are biologically productive but face severe ecological threats due to pollution, overexploitation of marine resources, unregulated tourism, and weak enforcement of conservation laws (Islam et al., 2021). The country's Marine Protected Areas (MPAs) were established to protect marine biodiversity; however, their effectiveness is often limited due to poor coordination, lack of capacity, and inadequate community engagement (Rahman, 2021).

2.2 Marine Megafauna and Conservation Challenges:

Marine megafauna—such as cetaceans, elasmobranchs (sharks and rays), and sea turtles—are vital indicators of ocean health and contribute to ecological balance. These species face threats from bycatch, habitat degradation, and marine traffic. Conservation strategies must consider speciesspecific data on habitat use, breeding grounds, and migration patterns. Despite the legal frameworks in place, enforcement remains a significant barrier due to overlapping mandates between agencies and inadequate marine crime monitoring systems (Hossain et al., 2022).

2.3 Coastal Community Livelihoods and Marine Resource Dependency:

Bangladesh's coastal districts are home to millions of people whose livelihoods are deeply tied to the sea. Fishing, tourism, and small-scale trade dominate these economies. However, unsustainable practices and environmental shocks (like cyclones and salinity intrusion) threaten both biodiversity and human well-being. Literature suggests that integrating climate-resilient livelihoods, such as eco-tourism and sustainable aquaculture, can reduce dependency on extractive marine practices and support conservation goals (Rahman, 2024). Public-private partnerships and gender-inclusive economic programs are essential for long-term sustainability.

2.4 Marine Pollution and Public Health Concerns:

Marine pollution—particularly from plastics, untreated sewage, ship waste, and industrial runoff—is a growing concern in the Bay of Bengal. Pollutants entering through rivers accumulate in coastal zones and MPAs, affecting not only marine ecosystems but also public health through contaminated seafood. Citizen science, open-access pollution portals, and educational campaigns have proven effective in mobilizing communities and increasing policy attention (Karmaker et al., 2024). Innovative solutions, including community-led monitoring and pollution tracking using digital tools, are increasingly being recognized as practical, low-cost approaches.

2.5 Coral Biodiversity of Saint Martin's Island:

Saint Martin’s coral ecosystem is home to 66 species of Scleractinian (hard) corals and 11 soft coral species (Gazi et al., 2020). These coral colonies provide essential habitat to 234 fish species, 186 mollusks, multiple species of echinoderms, and endangered marine turtles such as Olive Ridley and green turtles (Ahammed et al., 2016).

2.6 Coral Degradation and Climate Change:

Coral bleaching events have increased significantly since 1998, driven by a 0.45°C rise in sea surface temperature over the last four decades. This warming trend, combined with acidification and sedimentation, threatens the survival of coral communities and the biodiversity they support (Rahman, 2024).

2.7 Anthropogenic Threats:

Tourism-related pressure—including reef trampling, boat anchoring, and plastic waste—has directly reduced live coral coverage. Coral extraction and land-based pollution further weaken the reef system (Karmaker et al., 2024). Research highlights that coral area has declined by 0.9342 km^2 from 1980 to 2018 due to human interference (Gazi et al., 2020).

2.8 Research Gaps and Community Challenges:

There is a lack of recent coral mapping and long-term ecological monitoring, making it difficult to prioritize conservation zones. Existing co-management efforts suffer from low community participation, inequity, and lack of incentives for local fishers to engage in conservation (Rahman, 2021).

3.Material and methods

3.1 Study Area:

The internship activities were conducted across two key marine conservation contexts in Bangladesh:

Marine Protected Areas (MPAs) in the Bay of Bengal:

The project focused on five MPAs located along the southeastern and southwestern coasts of Bangladesh. These include areas within or near:

- Nijhum Dwip
- Sundarbans Reserve Forest
- Swatch of No Ground
- Teknaf Peninsula
- Saint Martin’s Island

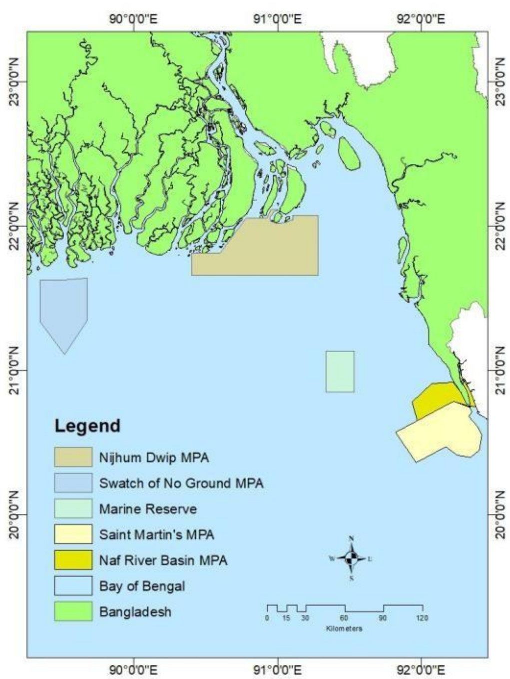

Figure 1 : 5 marine protected areas in the Bay of Bengal

These sites are ecologically significant for their high biodiversity, including populations of dolphins, sharks, rays, sea turtles, and diverse fish species. The MPAs are also socio-economically vital as they support the livelihoods of local fishing communities.

Saint Martin's Island:

This coral-rich island in the northeastern Bay of Bengal was a focal site for coral restoration planning. The island is the only coral-bearing zone in Bangladesh and is home to more than 60 coral species, as well as endangered sea turtles and reef-associated fish. The island faces growing threats from tourism pressure, climate change, and pollution, making it a priority area for coral vulnerability assessments and reef restoration initiatives.

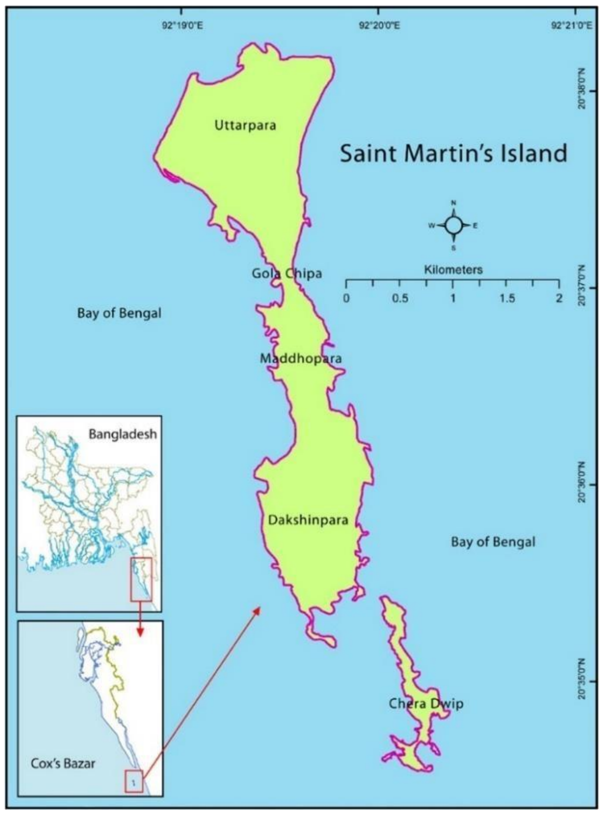

Figure 2: Saint Martin Island

3.2 . Methods

3.2.1 Water Quality Assessment

Sampling Locations: 5 Marine Protected Areas and Saint Martin's coral zone

- **CTD Profiler:** For measuring conductivity, temperature, and depth of seawater.

Figure 3: CTD Profiler

- **Dissolved Oxygen (DO) Meter** and **pH Meter:** For assessing water quality.

Figure 4: Do and PH meter

- **Niskin Water Sampler:** For collecting water samples at specific depths

Figure 5: Niskin Water Sampler

Plankton Net: Used to collect plankton samples from the water column for assessing marine biodiversity and ecosystem health.

Figure 6 : Plankton Net

Algae Torch : Use the algae torch to illuminate and study marine algae during night dives

Figure 7: Algae torch

3.2.2. Questionnaire Development: A semi-structured questionnaire was prepared covering:

- Livelihood activities and dependency on marine resources
- Knowledge of marine megafauna and MPA rules
- Experience with marine pollution and climate impacts
- Willingness to participate in conservation and alternative livelihoods

3.2.3. Target Groups: Fishermen, women in coastal households, youth, boat operators, local business owners, and local government representatives

3.2.4. Data Collection: Designed for in-person interviews and potential online deployment (if required)

3.2.5. Analysis: Quantitative responses will be analyzed using Excel and R; qualitative responses coded for thematic analysis

3.2.6. Stakeholder Mapping and Institutional Assessment

A stakeholder matrix was developed to map roles, influence, and engagement of:

- Government agencies (e.g., Forest Department, Department of Fisheries)
- NGOs and civil society groups
- Coastal communities and private sector actors

3.2.7. Coral Mapping and Index Planning

- Combined coral stressor layers (e.g., tourism, temperature rise, pollution) with coral health indicators
- Collaborated with DU, BFRI, and SUST labs to plan Coral Vulnerability Index (CVI)
- Outlined coral restoration zoning based on mapped habitat quality and threats

4. <u>Activities</u>

4.1.

Table 1: **Marine Conservation Activities and Intermediate Results**

IR Code	Objective	Key Focus Areas
IR 1.1	Knowledge on megafauna & habitats	Mapping, youth training, grants, rescue practices
IR 1.2	Evidence-based megafauna conservation	Policy review, law improvement ,tracking methods
IR 1.3	Reduce bycatch	Fisher training, encounter recording, rescue support
IR 1.4	Conservation financing	Public-private funding, ecoenterprises, livelihoods
IR 1.5	Marine pollution awareness	Pollution surveys, health risk analysis, knowledge portal
IR 2.1	Agency coordination	Joint workshops, protocols, marine forum
IR 2.2	Decision-maker engagement	Policy dialogue, workshops, advocacy
IR 2.3	MPA monitoring	Frameworks, community teams, tech tools
IR 2.4	Livelihood improvement	Training, eco-tourism, market linkage
IR 3.1	Marine crime prevention	Capacity building, legal tools, tracking
IR 3.2	Fish landing site monitoring	Site ID, guides, inspections

IR 3.3	Law enforcement technology	SMART patrols, data sharing
IR 3.4	Species trade compliance	Surveillance, prosecution, certification
IR 4.1	Community conservation actions	Campaigns,comanagement,volunteers
IR 4.2	Stakeholder participation	Analysis, partnerships, awareness surveys
IR 4.3	Youth & media involvement	Social media, education, journalism support

4.2 . Proposed Outcomes:

I. **Coral Vulnerability Index (CVI)**: The project will develop a CVI to identify areas with the highest vulnerability to degradation. This index will aid in targeted conservation efforts. II. **Assessment of Coral Cover**: A detailed survey of current coral coverage will be conducted to establish baseline data and track future restoration progress.

III. **Mitigation Framework**: A framework will be developed to reduce and mitigate coral destruction through community engagement, stricter regulations for tourism, and sustainable practices.

IV. **Coral Restoration**: Efforts will include lab-based coral nurturing and transplantation into suitable habitats to promote regrowth and biodiversity recovery.

5. Results and analysis

Coral Restoration Feasibility Study

Location: Saint Martin’s Island

Data Source: In-situ oceanographic sensor readings (Dec 2024)

Parameters Analyzed: Temperature, Salinity, DO, Oxygen Saturation, Chlorophyll-a

a) Temperature vs depth

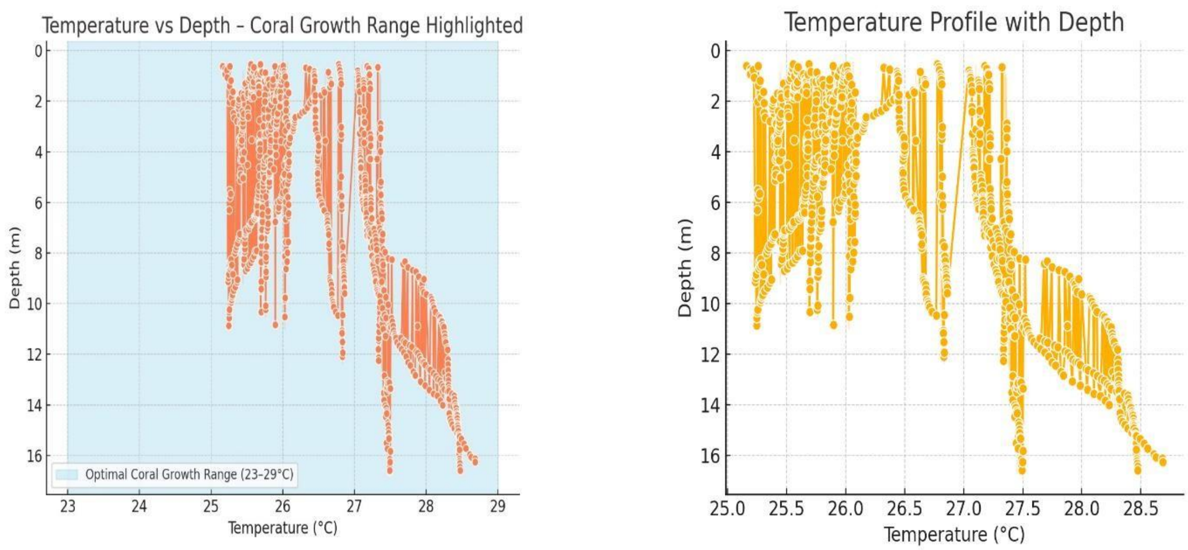

Figure 8 : Temperature vs depth

Surface temperatures range between 25–28.7°C. The upper 15 m of the water column lies entirely within the ideal range for coral growth (23–29°C). Conditions are highly favorable for coral health and symbiont algae photosynthesis.

b) Salinity vs Depth

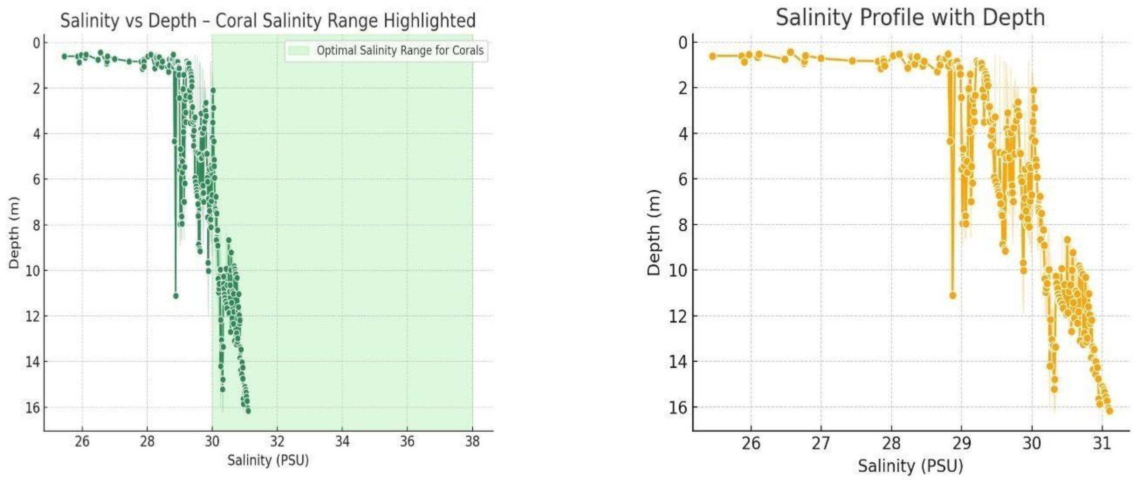

Figure 9 : Salinity vs Depth

Salinity remains stable between 29–31 PSU, slightly under the classic reef threshold (30–38 PSU). No sharp salinity drops, indicating no major freshwater runoff or estuarine stressors. Salinity conditions are within tolerance for most coral species.

c) Dissolved Oxygen (DO) vs Depth

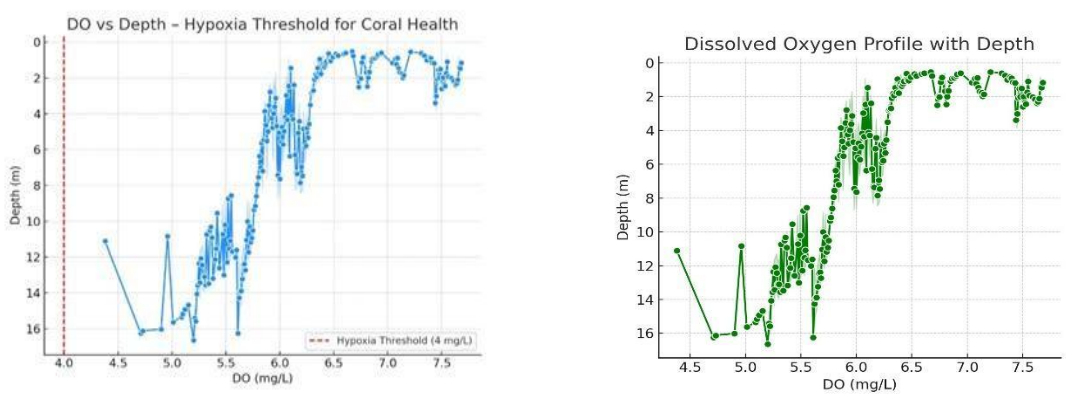

Figure 10 : DO vs Depth

DO ranges from 4.3–7.7 mg/L — all values are safely above hypoxia threshold (4 mg/L). Indicates active water circulation and biological support. No oxygen stress — highly supportive for coral and associated reef organisms.

d) Chlorophyll-a vs Depth

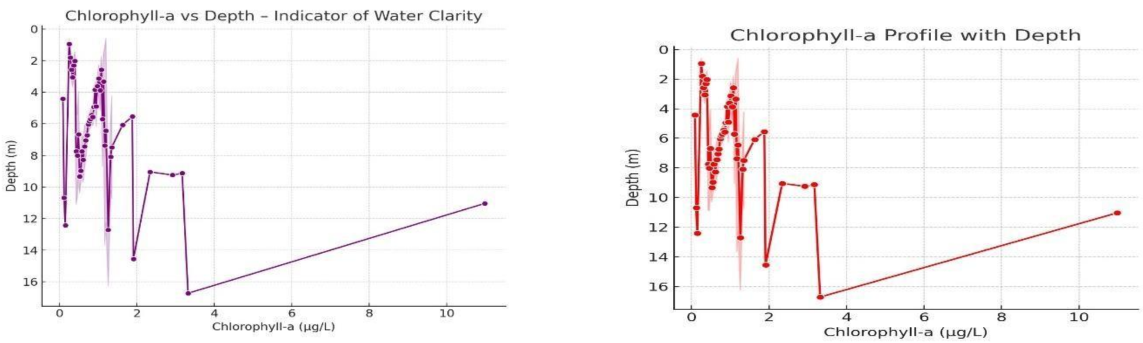

Figure 11: Chlorophyll-a vs Depth

Moderate Chlorophyll-a levels, typically below 1 µg/L, suggesting healthy water clarity and nutrient balance. No harmful algal bloom indicators. Sufficient light penetration for coral symbionts — a good sign for restoration.

e) Temperature vs DO (Colored by Salinity)

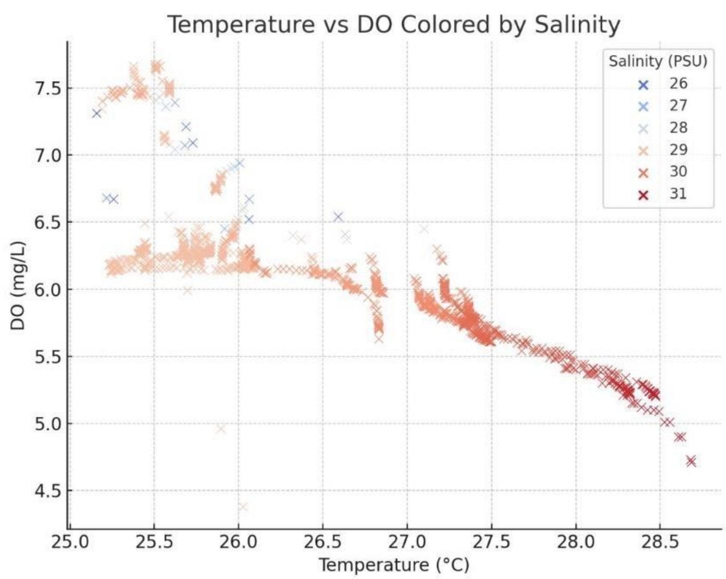

Figure 12: Temperature vs DO (Colored by Salinity)

Reveals a moderate positive relationship between temperature and DO in the coral-favorable salinity range. Warmer, shallower water with stable salinity supports both oxygenation and coral metabolism.

f) DO vs Depth (Colored by Temperature, Sized by Chl-a)

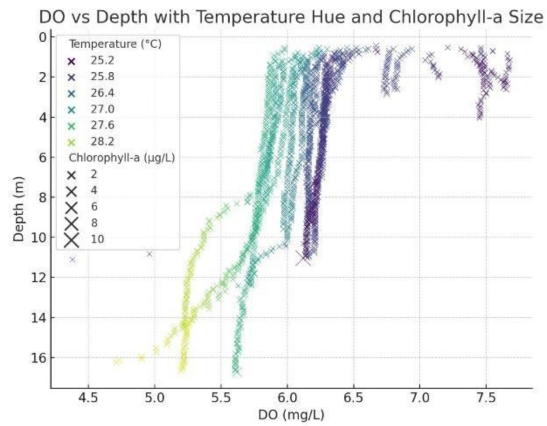

Figure 13 : DO vs Depth (Colored by Temperature, Sized by Chl-a)

Deeper water shows slight decline in DO, but still within safe limits. Larger chlorophyll dots near the surface show productive, oxygen-rich zones. Surface zone (0–10 m) is ideal for coral nursery placement.

g) Temperature vs Salinity (Colored by DO, Sized by Depth)

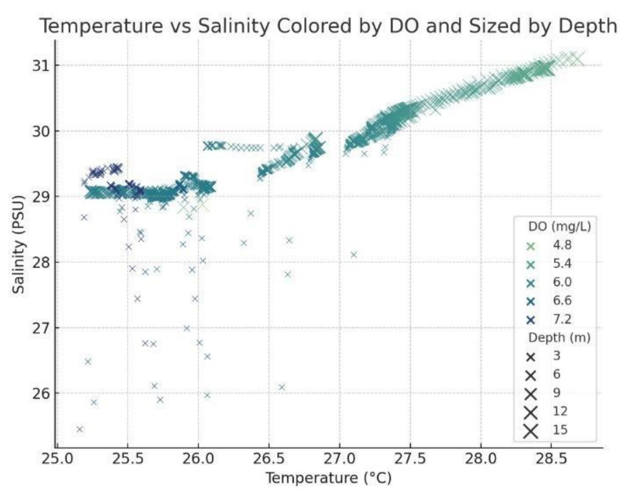

Figure 14: Temperature vs Salinity (Colored by DO, Sized by Depth)

Cluster of optimal coral conditions found around 26–27°C and 29–30 PSU, all with healthy DO levels. Confirms vertical and horizontal water mass consistency for restoration planning.

Table :2 Environmental Parameters for Coral Suitability

Parameter	Range	Feasibility Verdict
Temperature (°C)	25–28.7	Ideal for coral growth
Salinity (PSU)	29–31	Stable and suitable
DO (mg/L)	4.3–7.7	Safe, oxygen-rich
Chlorophyll-a (µg/L)	0.1–1.5 (mostly)	Healthy productivity
Depth (m)	0–10	Optimal zone for coral work

6. Discussion: Coral Restoration Feasibility in Saint Martin's Island

The environmental data collected from Saint Martin's Island during December 2024 reveals a highly favorable biophysical setting for coral restoration activities. The analysis of oceanographic parameters—temperature, salinity, dissolved oxygen (DO), and chlorophyll-a—demonstrates a stable, non-stressful habitat suitable for coral growth, photosynthesis, and overall reef recovery.

Sea surface temperatures ranged between 25–28.7°C, placing the upper 15 meters of the water column within the optimal thermal range for coral symbiont photosynthesis (23–29°C) (HoeghGuldberg, 1999). These values are particularly promising given the increasing concern over global coral bleaching thresholds, typically associated with prolonged exposure above 30–31°C. The absence of thermal anomalies during the sampling period indicates a low short-term bleaching risk, which is ideal for both nursery placement and coral transplantation efforts.

The measured salinity levels (29–31 PSU) were slightly below the classic reef tolerance threshold (30–38 PSU) but showed no sharp freshwater intrusion patterns, suggesting minimal estuarine influence or terrestrial runoff. According to Jokiel (2004), corals can tolerate a salinity range as low as 28 PSU provided it is consistent and free from fluctuations. The data supports the conclusion that salinity in Saint Martin's waters remains within tolerance limits, providing physiological stability for coral polyps and larvae settlement.

Dissolved oxygen levels ranged from 4.3 to 7.7 mg/L, comfortably above the hypoxia threshold of 4.0 mg/L. This indicates active water exchange, supporting aerobic respiration for coral animals and reef-associated fauna. Well-oxygenated water is also essential for nitrification processes, which help regulate nutrient levels within reef ecosystems (Fabricius, 2005). The data implies that the studied zone is not subject to stagnation or stratification, both of which are unfavorable for coral growth.

Chlorophyll-a concentrations were moderate (<1 μg/L), suggesting balanced primary productivity and sufficient light penetration to support photosynthetic activity of zooxanthellae. Unlike eutrophic waters, which often induce algal overgrowth and reduce coral fitness, the recorded chlorophyll levels fall within the range typical of healthy, oligotrophic reef environments (McCook, 1999). The absence of harmful algal bloom indicators further enhances the site's suitability for restoration.

Multi-parameter plots (e.g., DO vs Depth; Temperature vs Salinity) confirm that the upper 0–10 meters of the water column exhibits ideal conditions for coral nursery placement. These depths offer not only appropriate thermal and light conditions but also stable DO and salinity regimes, reducing physiological stress on transplanted corals.

7. <u>GIS Training Program Management</u>

During my internship at WildTeam Noazesh Center, I was responsible for managing the training session titled *"GIS in Research with QGIS."* My role involved coordinating the schedule, liaising with trainers and participants, and ensuring smooth delivery of the training content. This training focused on introducing QGIS software applications for research purposes, enhancing participants' skills in spatial data analysis and visualization. Through this experience, I gained valuable project management skills, improved my communication abilities, and deepened my understanding of GIS applications in environmental research.

8. Conclusion

My internship at Cosmos Foundation has been a profoundly enriching academic and professional experience, allowing me to explore real-world applications of marine conservation science. Over the course of the internship, I engaged in multidisciplinary research that combined field-based environmental monitoring, scientific proposal writing, and stakeholder-oriented planning for biodiversity protection across Bangladesh’s coastal and marine areas.

A central highlight of my internship was conducting a coral restoration feasibility study in Saint Martin’s Island. The in-situ oceanographic data I analyzed—covering parameters such as temperature, salinity, dissolved oxygen, and chlorophyll-a—provided strong evidence that the region’s shallow waters are ecologically suitable for coral nursery establishment and reef rehabilitation. The scientific process of evaluating coral habitat conditions has not only deepened my technical understanding but also strengthened my data analysis and scientific writing skills.

Additionally, I contributed to the preparation of a broader conservation proposal that addressed threats to marine megafauna, pollution in Marine Protected Areas (MPAs), and climate-resilient community planning. This involved synthesizing literature, analyzing policies, and drafting outcomes and activities that align with national priorities and international conservation goals.

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